Quotients and Remainders with Models

Reading a quotient off a model, deciding what the remainder means, and estimating first

Directions:

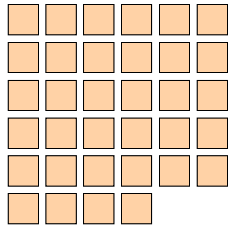
- Every division here can be written as (total) = (groups) $\times$ (size of a group) + (remainder), and the remainder is always smaller than the group size.
- Where a model is drawn, mark it up: circle groups, cross off tiles, label what is left over.
- Read each question carefully — sometimes the answer is the quotient, sometimes the remainder, and sometimes one more than the quotient.

1. The counters below are shared into equal groups of 5. Circle the groups.

(a) How many full groups of 5 are there? _________

(b) How many counters are left over? _________

2. A tile pattern is laid out in rows of 6 tiles. There are 34 tiles in the pattern.



(a) How many complete rows of 6 are there? _________

(b) How many tiles are left over, not in a complete row? _________

3. A class of 38 students is travelling to a museum. Each van seats 8 students.

(a) How many vans are needed so that every student has a seat?

Answer: __________________ vans

(b) Explain why the leftover students change the answer you would get from dividing alone.

4. A baker packs 45 cookies into boxes that hold 6 cookies each. Only full boxes are sold. How many full boxes can the baker sell?

Answer: __________________

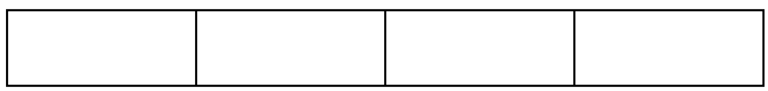
5. A gardener shares 356 bulbs equally among 8 beds.

(a) Round 356 to the nearest hundred. _________

(b) Use that to estimate how many bulbs go in each bed. _________

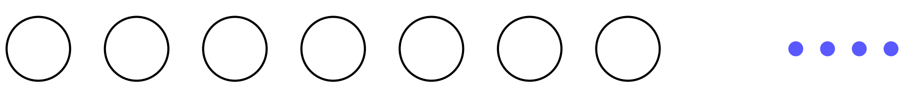
(c) Now divide exactly: each bed gets _________ bulbs, with _________ bulbs left over.

6. The bar below stands for 96 counters, split into 4 equal parts. How many counters are in each part?



Answer: _________________

7. The model below shows 60 counters put into 7 equal groups, with 4 counters left over. How many counters are in each group?



Answer: _________________

8. A collector puts 52 marbles into bags of 9. How many marbles are left over after every full bag is filled?

Answer: ________________

9. A printer produces 1,235 leaflets, packed equally into 6 cartons.

(a) Round 1,235 to the nearest hundred. ________

(b) Use that to estimate how many leaflets go in each carton. ________

(c) Now divide exactly: each carton holds ________ leaflets, with ________ left over.

10. 83 pencils are shared among 6 friends.

(a) How many pencils does each friend get? ________

(b) How many pencils are left over? ________

(c) Now suppose the same 83 pencils are packed into boxes that hold 6 pencils. How many boxes are needed to hold every pencil? ________

(d) The same division gave two different answers in (a) and (c). Explain why.